Kroki

Creates **diagrams** from **textual** descriptions!

Kroki provides a unified API with support for BlockDiag (BlockDiag, SeqDiag, ActDiag, NwDiag, PacketDiag, RackDiag), BPMN, Bytefield, C4 (with PlantUML), D2, DBML, Ditaa, Erd, Excalidraw, GraphViz, Mermaid, Nomnoml, Pikchr, PlantUML, Structurizr, SvgBob, Symbolator, TikZ, UMLet, Vega, Vega-Lite, WaveDrom, WireViz... and more to come!

> 📢 **We're planning the future of Kroki, and we need your insights!**
> Please take a moment to fill out our short questionnaire – your feedback is crucial in helping us grow.

Carbon
What makes you click on a dev tool ad?
Honest answers only. 3-min survey.
ADS VIA CARBON

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#Features

Ready to use
Diagrams libraries are written in a variety of languages: Haskell, Python, JavaScript, Go, PHP, Java... some also have C bindings. Trust us, you have better things to do than install all the requirements to use them. Get started in no time!

Simple
Kroki provides a unified API for all the diagram libraries. Learn once use diagrams anywhere!

Free & Open source
All the code is available on GitHub and our goal is to provide Kroki as a free service.

Fast
Built using a modern architecture, Kroki offers great performance.

#Try

Diagram

PlantUML

```
Vista --> User: 12. Redirigir al Dashboard Principal (SAGA
deactivate Vista

== Flujo Alterno: Bloqueo de Cuenta por 3 Intentos Fallido

User -> Vista: 1. Ingresar credenciales incorrectas\n(Inte
activate Vista
Vista -> Ctrl: 2. autenticarUsuario(usuario, password)
activate Ctrl
Ctrl -> Serv: 3. validarCredenciales(usuario, password)
activate Serv
Serv -> DB: 4. buscarUsuario(usuario)
activate DB
DB --> Serv: 5. return (cuentaBloqueada=false, intentosFal
Serv -> DB: 6. registrarIntentoFallidoYBloquear(usuario)
note right of DB: Actualiza cuentaBloqueada = true\nen la
DB --> Serv: 7. cuenta modificada a bloqueada
deactivate DB
Serv --> Ctrl: 8. return (error: Cuenta Bloqueada por Segu
deactivate Serv
Ctrl --> Vista: 9. return (Acceso Denegado)
deactivate Ctrl
Vista --> User: 10. Mostrar Alerta Roja\n("Cuenta bloquead
deactivate Vista

@enduml
```

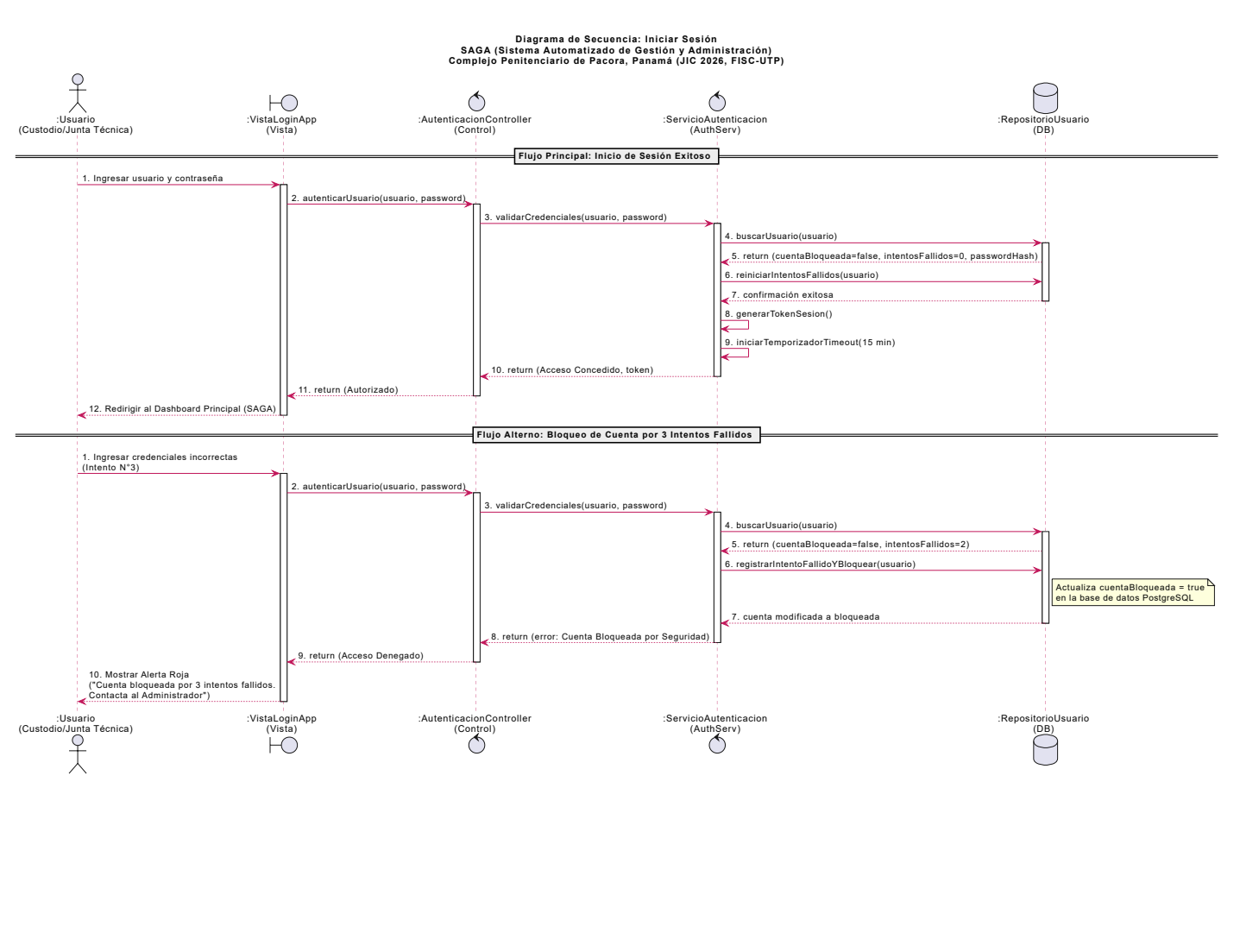


Diagrama de Secuencia: Iniciar Sesión
SAGA (Sistema Automatizado de Gestión y Administración)
Complejo Penitenciario de Pacora, Panamá (JIC 2026, FISC-UTP)
:Usuario (Custodio/Junta Técnica)
:VistaLoginApp (Vista)
:AutenticacionController (Control)
:ServicioAutenticacion (AuthServ)
:RepositorioUsuario (DB)
Flujo Principal: Inicio de Sesión Exitoso
1. Ingresar usuario y contraseña
2. autenticarUsuario(usuario, password)
3. validarCredenciales(usuario, password)
4. buscarUsuario(usuario)
5. return (cuentaBloqueada=false, intentosFallidos=0, passwordHash)
6. reiniciarIntentosFallidos(usuario)
7. confirmación exitosa
8. generarTokenSesion()
9. iniciarTemporizadorTimeout(15 min)
10. return (Acceso Concedido, token)
11. return (Autorizado)
12. Redirigir al Dashboard Principal (SAGA)
Flujo Alterno: Bloqueo de Cuenta por 3 Intentos Fallidos
1. Ingresar credenciales incorrectas (Intento N°3)
2. autenticarUsuario(usuario, password)
3. validarCredenciales(usuario, password)
4. buscarUsuario(usuario)
5. return (cuentaBloqueada=false, intentosFallidos=2)
6. registrarIntentoFallidoYBloquear(usuario)
Actualiza cuentaBloqueada = true en la base de datos PostgreSQL
7. cuenta modificada a bloqueada
8. return (error: Cuenta Bloqueada por Seguridad)
9. return (Acceso Denegado)
10. Mostrar Alerta Roja ("Cuenta bloqueada por 3 intentos fallidos. Contacta al Administrador")

```
GET https://kroki.io/plantuml/svg/eNrtVt1u4kYUvvdTHKUXBYlQIJsti0S1YMI2q7SigVSqlJuDfYBJxjN0ZpztpurD9HIvKrXqG5QX65m
```

Copy

#Examples

Looking for inspiration? Visit the examples page.

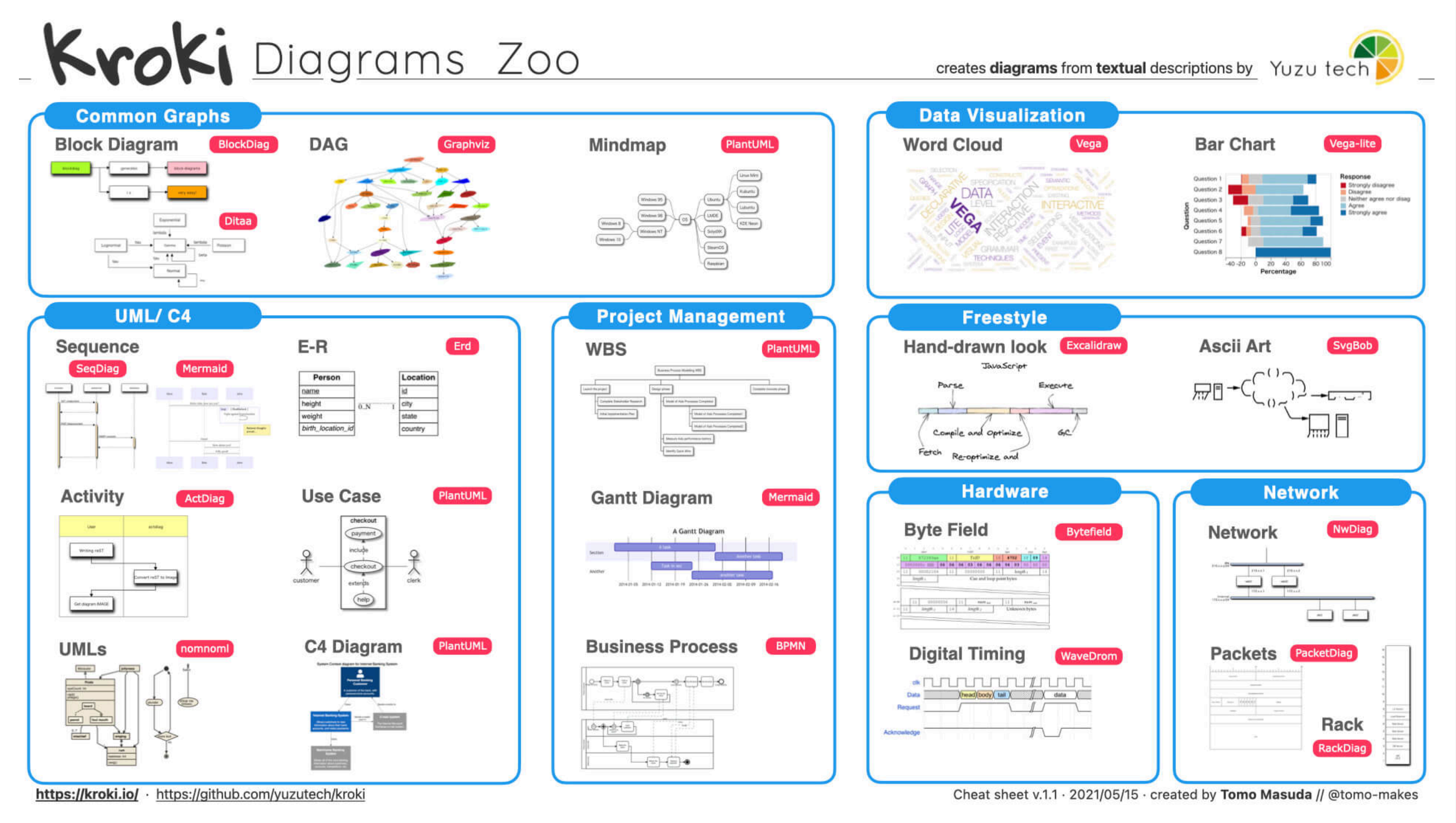
#Diagram types

The following diagram types and output formats are available:

Diagram Type	png	svg	jpeg	pdf	txt	base64
BlockDiag	✓	✓		✓		
BPMN		✓				
Bytefield		✓				
SeqDiag	✓	✓		✓		
ActDiag	✓	✓		✓		
NwDiag	✓	✓		✓		
PacketDiag	✓	✓		✓		
RackDiag	✓	✓		✓		
C4 with PlantUML	✓	✓		✓	✓	✓
D2		✓				
DBML		✓				
Ditaa	✓	✓				
Erd	✓	✓	✓	✓		
Excalidraw		✓				
GraphViz	✓	✓	✓	✓		
Mermaid	✓	✓				
Nomnoml		✓				
Pikchr		✓				
PlantUML	✓	✓		✓	✓	✓
Structurizr	✓	✓		✓	✓	✓
Svgbob		✓				
Symbolator		✓				
TikZ	✓	✓	✓	✓		
UMlet	✓	✓	✓			
Vega	✓	✓		✓		
Vega-Lite	✓	✓		✓		
WaveDrom		✓				
WireViz	✓	✓				

You don't see your favorite diagram tool in this list, please let us know 👋 hello@kroki.io.

#Cheat sheet


Kroki Diagrams Zoo
creates diagrams from textual descriptions by Yuzu tech
Common Graphs
Block Diagram
BlockDiag
DAG
Graphviz
Mindmap
PlantUML
Ditaa
Data Visualization
Word Cloud
Vega
Bar Chart
Vega-lite
UML/ C4
Sequence
SeqDiag
Mermaid
E-R
Erd
Activity
ActDiag
Use Case
PlantUML
UMLs
nomnoml
C4 Diagram
PlantUML
Project Management
WBS
PlantUML
Gantt Diagram
Mermaid
Business Process
BPMN
Freestyle
Hand-drawn look
Excalidraw
Ascii Art
SvgBob
Hardware
Byte Field
Bytefield
Digital Timing
WaveDrom
Network
Network
NwDiag
Packets
PacketDiag
Rack
RackDiag
https://kroki.io/ · https://github.com/yuzutech/kroki
Cheat sheet v.1.1 · 2021/05/15 · created by Tomo Masuda // @tomo-makes

📌 Download cheat sheet as PDF

#Install

Kroki is available as a Free Service and as a Self-Managed instance.

Self-Managed

Select this option if you want to download and install Kroki on your own infrastructure or in a cloud environment.

> Please note that Kroki Self-Managed requires Linux experience to install.

Please refer to the documentation to install Kroki.

Free service

Thanks to Exoscale, which has kindly offered two servers, we can provide Kroki as a *free service*!
However, some parts of the infrastructure are still host on other providers, so **we are still looking for sponsors to cover all the expenses.**

If you are interested, please 👋 contact us.

#How

Kroki provides an HTTP API to create diagrams from textual descriptions. Kroki handles both `GET` and `POST` requests. When using `GET` requests, your diagram must be encoded in the URL using a deflate + base64 algorithm. But don't worry, if you're not familiar with deflate or base64 (or if you don't want to use them), you can also send your diagram as plain text using `POST` requests (see below).

Let's take an example with a *GraphViz* "Hello World":

`hello.dot`

```
digraph G {Hello->World}
```

Here, we are using a Python one-liner to encode our diagram using deflate + base64:

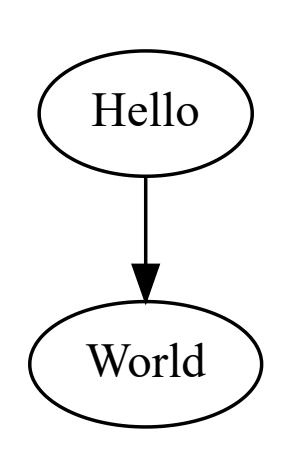
```
cat hello.dot | python -c "import sys; import base64; import zlib; print(base64.urlsafe_b64encode(zlib.compress(sys.stdin
```

> In the documentation, we provide code examples that demonstrate how to encode a diagram in Node.js, JavaScript, Java, Python and Go.

The above command will return a value that you can copy and paste at the end of the URL:

```
GET /graphviz/svg/eNpLyUwvSizIUHBXqPZIzcnJ17ULzy_KSanlAgB1EAjQ
```

And here's the result:


Hello
World

Please note that the encoding process is lossless. As a matter of fact, you can also use a Python one-liner to decode an encoded diagram:

```
echo "eNpLyUwvSizIUHBXqPZIzcnJ17ULzy_KSanlAgB1EAjQ" | python -c "import sys; import base64; import zlib; print(zlib.decom
```

The above command will return:

```
digraph G {Hello->World}
```

#POST requests

You can also call Kroki with `POST`:

```
POST /
```

```
{
  "diagram_source": "digraph G {Hello->World}",
  "diagram_type": "graphviz",
  "output_format": "svg"
}
```

In this case, you don't need to encode your diagram.

It's also possible to send your diagram as plain text using the `Content-Type` header. The output format will be specified using the `Accept` header and the diagram source will be sent as the request body:

```
POST /graphviz
```

```
Accept: image/svg+xml
Content-Type: text/plain

digraph G {
  Hello->World
}
```

You can also define the output format in the URL if you don't want to add an `Accept` header:

```
POST /graphviz/svg
```

```
Content-Type: text/plain

digraph G {
  Hello->World
}
```

The same concept applies when sending the diagram as `JSON`:

```
POST /graphviz/svg
```

```
{
  "diagram_source": "digraph G {Hello->World}"
}
```

Please note that you can interact with the API using **any HTTP client**.
If you want to learn more, head to our documentation. In particular, check out the Usage section to find out how to send requests to the Kroki API using:

- *cURL* and *HTTPie*
- Kroki CLI

#Sponsors

Thanks to our sponsors listed below, we are able to provide Kroki as a free service available to everyone.

EXOSCALE
A solid European cloud hosting alternative.

Terrastruct
The diagramming tool for developers.

ZULIP
An open-source modern team chat app designed to keep both live and asynchronous conversations organized.

UTOOLITY
Utilities and tools to integrate Amazon Web Services (AWS) with Atlassian products.

Sponsors play a critical role, without them, we won't be able to provide Kroki as a free service.
To make Kroki sustainable, we need your support. If you want to become a sponsor, please 👋 contact us.

#About

Kroki is a free service built and maintained by Yuzu tech and supported by Exoscale.

Kroki is an open source project licensed under the MIT license.
This means that Kroki is never going away; anyone can host this service with minimal resource requirements.